Low Temperature Generation and Measurements

- Introduction
- Why low temperature measurements are required in Physics ?
- Basics Introduction About Cryogens
- Brief Detail about Helium
- Details about Cryostats

What is temperature?

- On the **microscopic scale**, temperature is defined as the average kinetic energy of a particle in the system per degree of freedom. On the **macroscopic scale**, temperature is the unique physical property that determines the direction of heat flow between two objects placed in thermal contact.

- Internal energy, *U*, and entropy, σ are the fundamental parameters describing any system. The temperature is defined as

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V} \qquad \frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U} \right)_{N,V}$$

$k_B T = \tau$ - energy units

$\sigma = \frac{S}{k_B}$ - dimensionless (pure combinatorics)

convenient dimensionless form

- Boltzmann constant $k_B = R/N_A$

R = universal gas constant = 9.314 J mol^{-1} K^{-1}

N_A = Avogadro number $6.022 \times 10^{23}\ \mathrm{mol}^{-1}$

- A way to reconcile the mathematical stat mech with the observable temperature and the universal gas law

$$pV = nRT$$

- Thermal fluctuations – randomizing force --- $k_B T$

The Absolute Zero

- At absolute zero the entropy is minimal and the system is in its ground state. This includes **quantum mechanical zero-point energy.**
- At T=0, the molecular motion **does not cease** but does not have enough energy to be transferred to other systems. At T=0 molecular energy is minimal.

Why low temperature measurements are required in Physics ?

- When temperatures are lowered, the thermal energy of the system eventually becomes comparable to the ground state energy and the quantum-mechanical nature of the system becomes most apparent. For this reason, low temperatures are widely used in the study of quantum phenomena in condensed matter.

- At low temperature observed effects are free from thermal vibrations, it helps us to achieve pristine properties of matters

- At low temperature, we can observe new phenomenon like- Superconductivity.

- Low-temperature studies of matter can be used in order to learn more about how materials behave at much higher temperatures. For example, semiconductor devices often work at temperatures above room temperature. However, by cooling them down to remove the crystal lattice vibrations, the effects of impurities etc. on the electronic properties can be studied independently. The vibrations can be put back in a controlled manner by warming the sample slightly, so that their effect can be studied in more detail.

- It also used to reduce noise of electronics devices for ultra precision instruments like radio telescopes.

What temperature do we call a low temperature?

It is relative. For us, it is the line between high and low temperature will be drawn according to the technology required to achieve and work at those temperatures. Low temperatures will be those that require the use of liquefied gases with boiling points near to or below the boiling point of liquid nitrogen to be reached, or those temperatures lower than can be obtained by using conventional refrigeration techniques. This technology is called *cryogenics.*

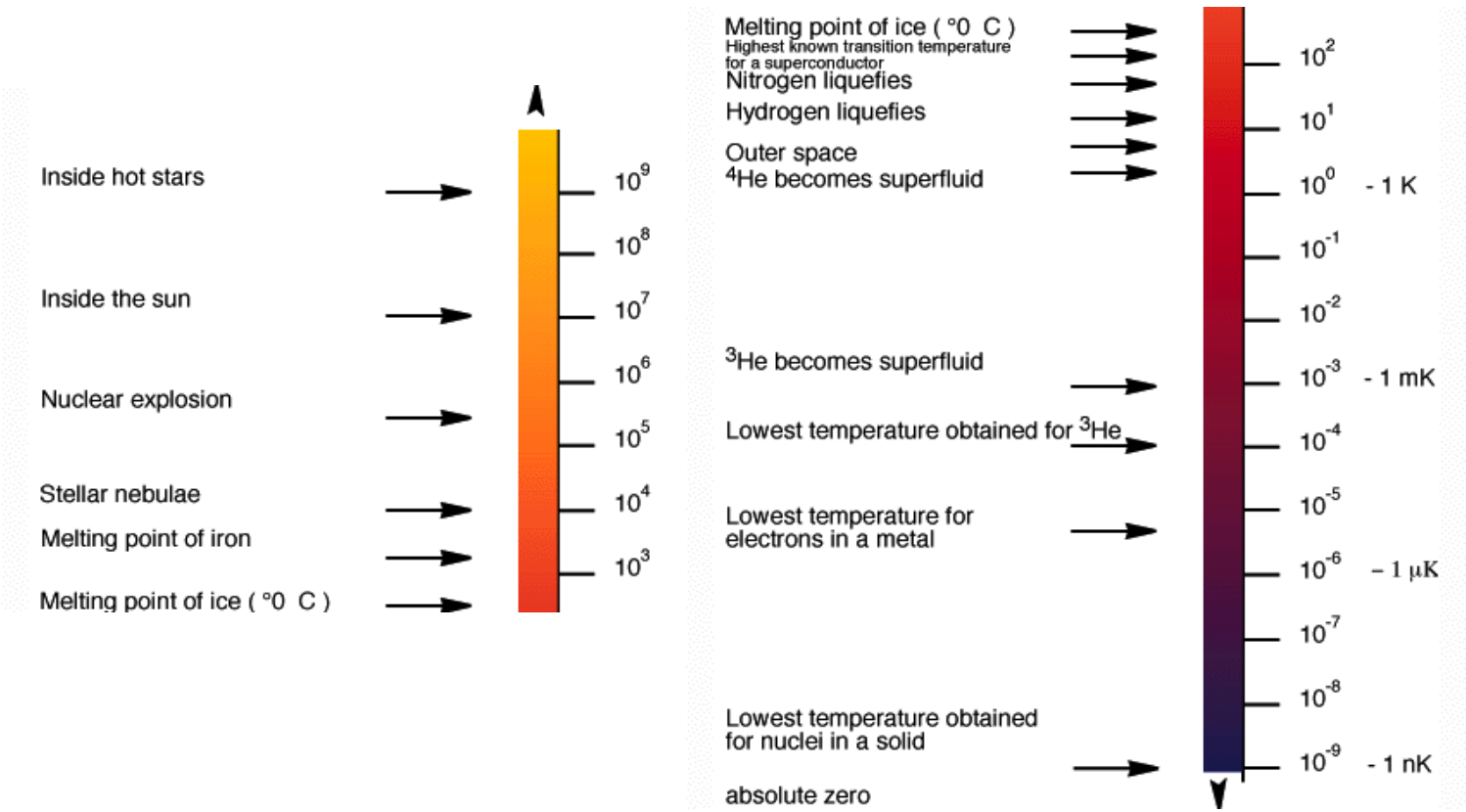
Some typical low temperatures

	Temperature	
	Celsius (°C)	Absolute (K)
Tropics	45	318
Human body	37	310
Room temperature	20	293
Ice point	0	273
Salt + water (cryogen)	-18	255
Antarctic winter	-50	223
Solid carbon dioxide	-78	195
Liquid oxygen	-183	90
Liquid nitrogen	-196	77
Liquid helium	-269	4
Absolute zero	-273	0

Characteristic temperatures of low-energy phenomena

Phenomenon	Temperature
Debye temperature of metals	few 100 K
High-temperature superconductors	~ 100 K
Low-temperature superconductors	~ 10 K
Intrinsic transport properties of metals	< 10 K
Cryopumping	few K
Cosmic microwave background	2.7 K
Superfluid helium 4	2.2 K
Bolometers for cosmic radiation	< 1 K
Low-density atomic Bose-Einstein condensates	~ μK

Temperature scales



Superfluidity in helium

The two isotopes of helium, ^{3}He and ^{4}He, have the lowest normal boiling points of all known substances

3.19K and 4.21K respectively

Under their saturated vapour pressures they remain liquid down to the very lowest temperatures

To produce a solid phase requires application of rather high pressures (approximately 25 atmospheres)

This is because of (a) the low mass of He

(b) weak interatomic forces between He atoms

(a) gives rise to large zero point energy:

assume a He atom is confined within a sphere of radius R bounded by other atoms - from the Heisenberg Uncertainty Principle we have

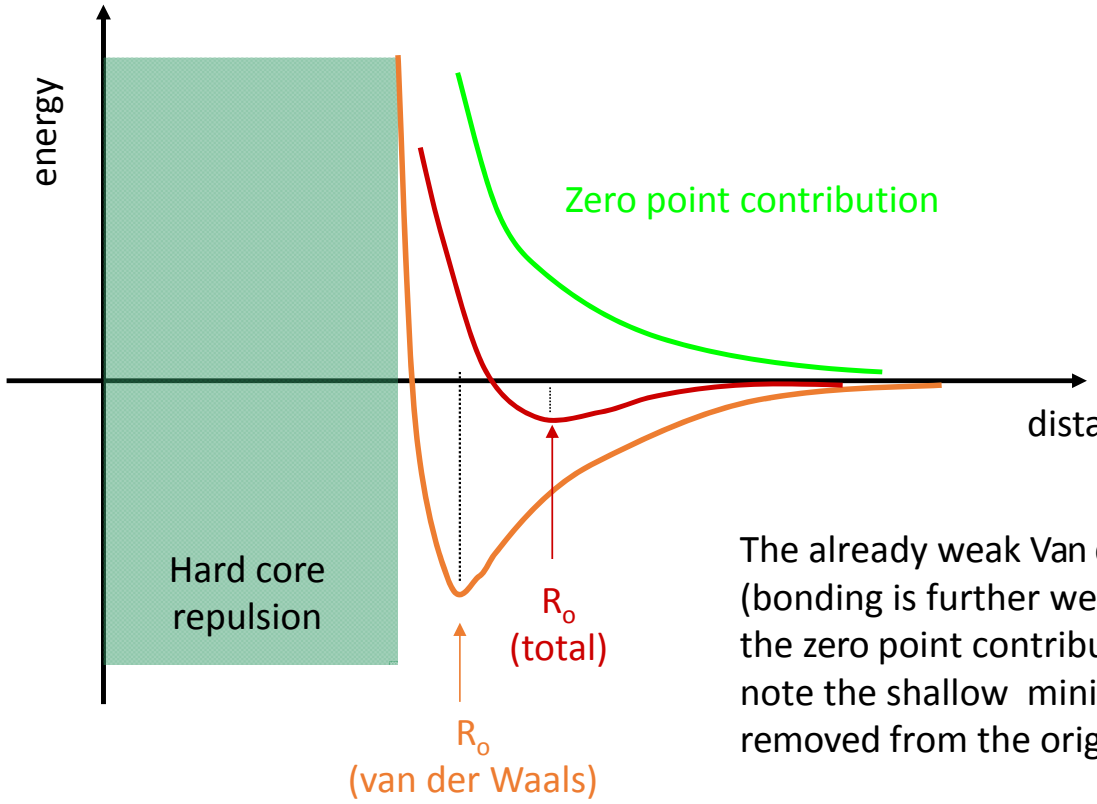
$$\Delta p \sim h/R$$

and for the energy

$$E_o = \frac{(\Delta p)^2}{2m_4} \sim \frac{h^2}{2m_4 R^2}$$

and this is the zero point correction to the total energy

Zero point effects



The already weak Van der Waals (bonding is further weakened by the zero point contribution - note the shallow minimum well removed from the origin

Basic properties of helium

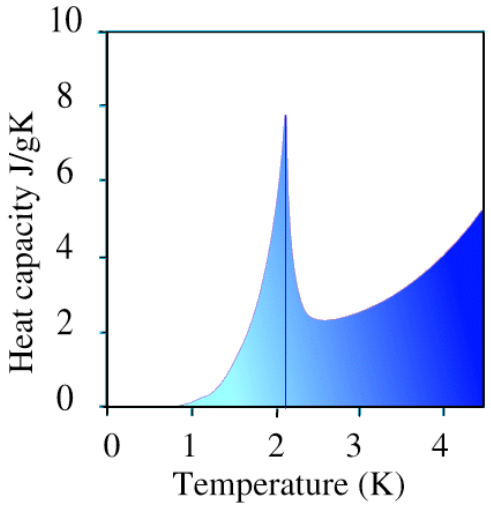
Immediately below their respective boiling points both ^{3}He and ^{4}He behave like ordinary liquids with a small viscosity

Their thermal and kinetic properties in this regime can be described with moderate success by a simple kinetic theory of gases

At low temperatures both show marked deviations from quasi-classical behaviour

In liquid ^{4}He the change in behaviour is marked by a characteristic anomaly in the heat capacity at 2.17K - the so called λ–transition because of its shape

Below this temperature the 4He liquid becomes superfluid -eg it can flow through capillaries with no viscosity



^{3}He

From its boiling point down to approximately 1K ^{3}He behaves like ^{4}He close to its boiling point

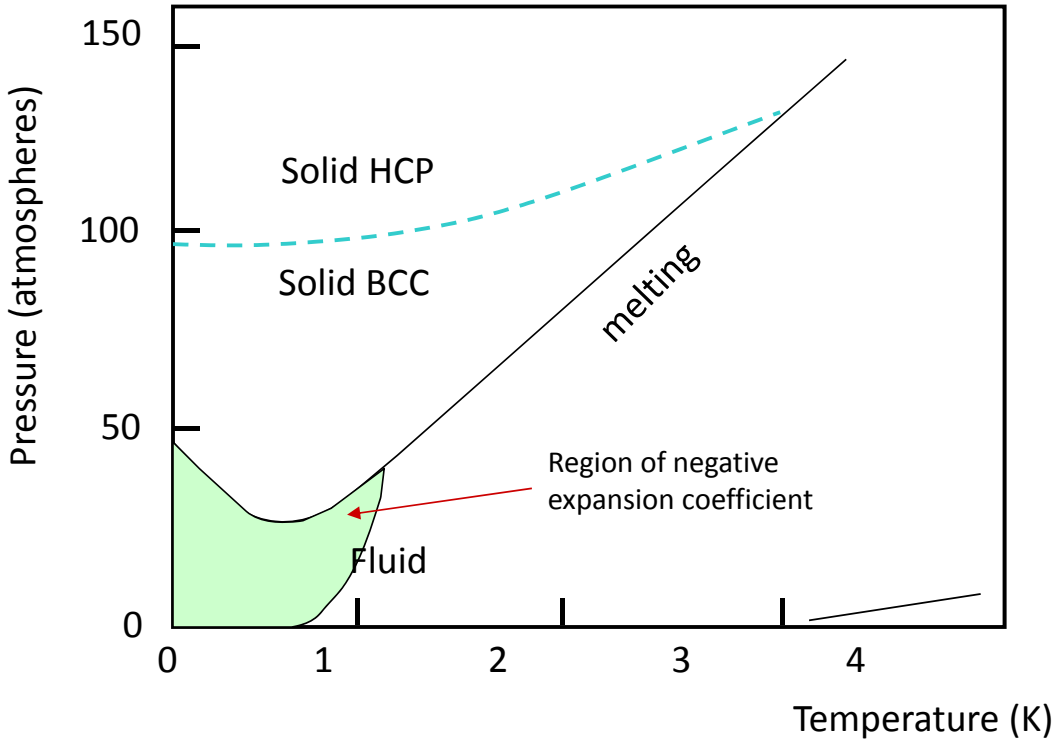
Between 1K and 0.003K ^{3}He behaves differently both to ^{4}He and quasi-classical expectations

At 2.6mK ^{3}He, like ^{4}He, also undergoes a transition to a superfluid state, signalled by an abrupt increase in the heat capacity

It should be noted that while ^{4}He is governed by Bose statistics, ^{3}He is governed by Fermi statistics

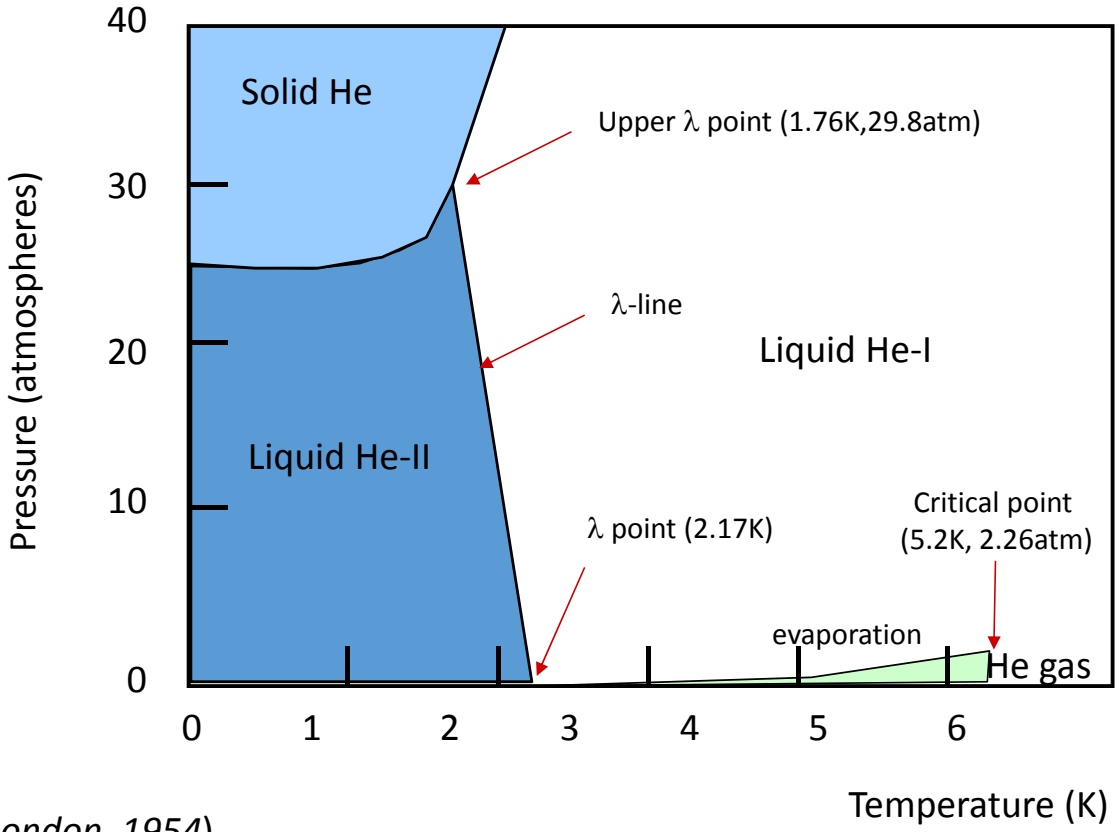
The superfluid transition in ^{3}He is associated with pairing of ^{3}He atoms, the resulting pairs then behaving as bosons

Phase diagram of ^{3}He



(after Grilly and Mills, 1959)

Phase diagram of ^{4}He



(*after London, 1954*)

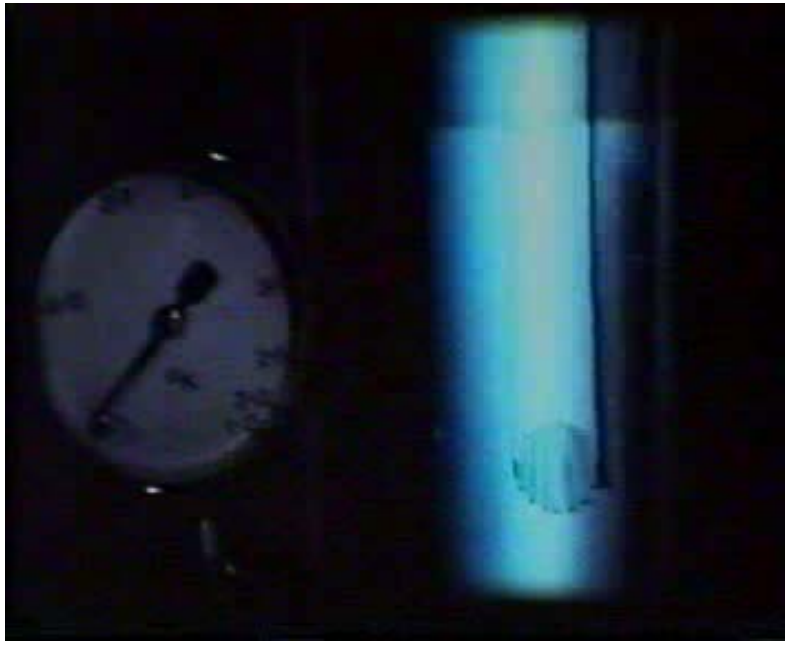
More About ^{4}He

Liquid He-II

The λ-transition marks an abrupt boundary between He-I and He-II

Above T_λ bubbles of vapour form within the bulk of the liquid and the whole liquid is agitated by the bubbles rising to the surface

Below T_λ the liquid becomes calm and still
- this is because it is almost impossible to set up an appreciable temperature gradient because heat transport takes place so readily, hence all evaporation takes place only at the free surface.



The λ-transition is sharp to within a mK.

Superfluid Video:

https://www.youtube.com/watch?v=2Z6UJbwxBZI https://www.youtube.com/watch?v=9FudzqfpLLs

https://www.youtube.com/watch?v=dLcwmMGCfU8

More About ^{4}He

He-4 nucleus has no spin, Boson

No solid phase due to:

weak van derWaals inter-atomic interactions, E_{pot} is low

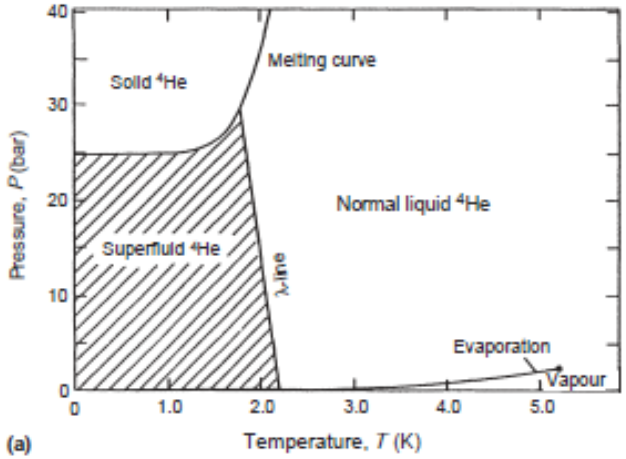
Large quantum mechanical zero-point energy $E_0=h^2/8ma^2$

due to small mass, E_{kin} is high

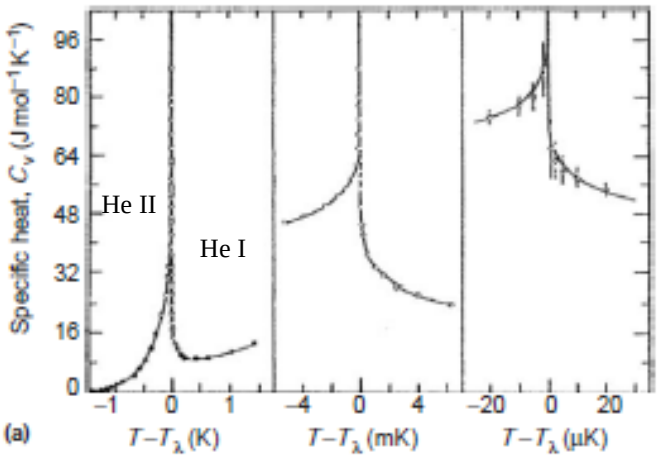
Bose-Einstein condensate instead of a solid

Quantum liquids, ratio $\lambda=E_{kin}/E_{pot}$ He4 - $\lambda=2.64$ He3 $\lambda=3.05$

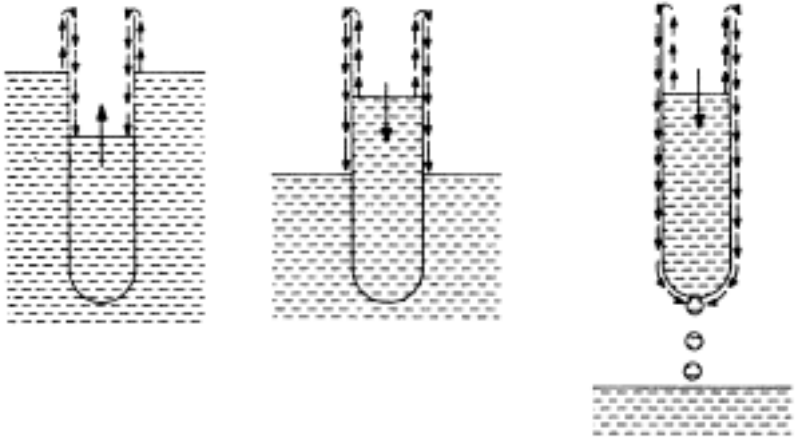
Amplitude of vibrations about 1/3 of interatomic space



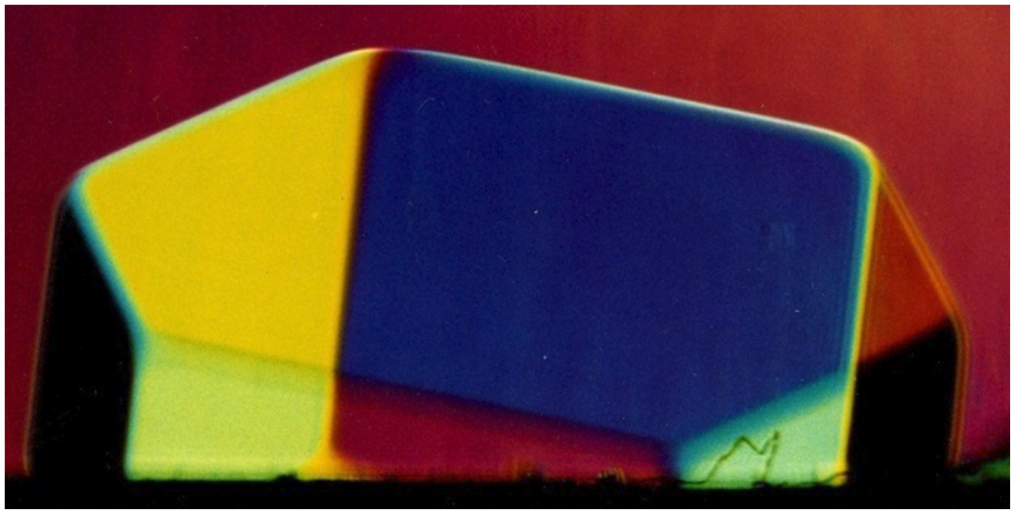
Heat Capacity – Lambda Point 2.1768 K

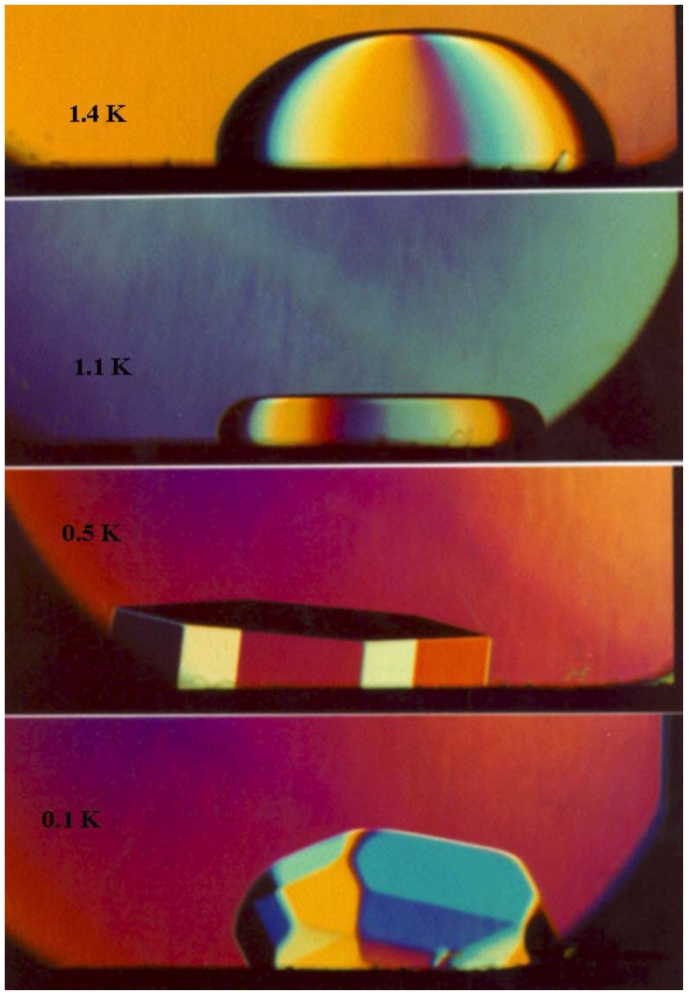


Superfluid



More About ^4He





Solid He 4 Crystals at different temperature. Depending upon temperature and presussure it can be crystalised in different crystal structures

Cooling Power of Cryogens:

Normal boiling point and latent heat of fluids

- **Water : 373 K 2256 kJ/kg**
- **Ethylene : 169 K 481 kJ/kg**
- **Krypton : 120 K 116 kJ/kg**
- **Methane : 111 K 512 kJ/kg**
- **Xenon : 110 K 99 kJ/kg**
- **Oxygen : 90 K : 213 kJ/kg**
- **Argon 87 K : 162 kJ/kg**
- **Nitrogen 77 K : 199 kJ/kg**
- **Neon 27 K : 86 kJ/kg**
- **Hydrogen 20 K 443 kJ/kg**
- **Helium : 4.2 K 21 kJ/kg**

Amount of cryoliquids (in liters) to cool 1 kg of metal if only latent heat (latent heat + enthalpy of the gas) is used.

Cryoliquid	Temperature change (K)	Al	SS	Cu
N_2	300 - 77	1.0 (0.63)	0.53 (0.33)	0.46 (0.28)
^4He	77 – 4.2	3.2 (0.2)	1.4 (0.1)	2.2 (0.16)
^4He	300 – 4.2	66 (1.6)	34 (0.8)	32 (0.8)

Use LN2 to precool the equipment (caution and patience needed) Make use of the enthalpy of the cold He gas (very slow initial transfer, end of the transfer tube close to the bottom)

What is lowest Temperature we can achieve with He 3 & He4

Temperature Reduced as we reduce pressure.

Cooling power proportional to vapor pressure Q~ P(T) ~exp(-1/T)

Exponentially small at low T

We can get by pumping
on He-4 T~1K
He-3 T~0.26 K

